

Vaibhav Kandpal

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Education

IIT Roorkee

Professional Certificate in Data Science & AI (6-month intensive)

Feb 2026 – Present

Remote

Dr. Akhilesh Das Gupta Institute of Professional Studies (GGSIPU)

Bachelor of Technology in Computer Science and Engineering

Aug 2024 – May 2028 (Expected)

New Delhi, India

Awards & Achievements

- NASA Space Apps Challenge 2024: Top 5 in India (823 competing teams); built OrbitAir — real-time air quality forecasting system integrating TEMPO satellite, EPA, OpenAQ & NOAA feeds. github.com/Vaibhav20k/ORBIT-AIR
- Smart India Hackathon 2024 — Participant
- Deloitte Australia Data Analytics Job Simulation — Forage certified (Jul 2025)

Projects

RepoLens Portfolio | React, FastAPI, PostgreSQL, Docker, GitHub Actions

github.com/Vaibhav20k/Repolens-portfolio

- Built automated GitHub repo-ingestion pipeline parsing READMEs, commit history & language stats via REST API — eliminated manual project entry for 100+ repos; live at vaibhavs.indevs.in.
- Cut AI response latency to p95 ~900ms by integrating OpenRouter (NVIDIA LLM) with streaming SSE, replacing a synchronous REST pattern and enabling real-time Portfolio Copilot chat.
- Achieved zero-downtime deployments via GitHub Actions CI/CD with Pytest smoke tests and automated Vercel preview builds on every pull request.
- Secured sessions with JWT refresh-token rotation and OAuth 2.0 GitHub sign-in; implemented role-based access control protecting all portfolio edit endpoints.

OrbitAir: AI-Based Air Quality Forecasting System | Python, React, TimescaleDB, FastAPI, Docker

github.com/Vaibhav20k/ORBIT-AIR

- Engineered multi-source ETL pipeline fusing TEMPO satellite, EPA, OpenAQ & NOAA feeds into TimescaleDB via async batch processing — reduced pipeline runtime from 8 min to under 2 min (4× throughput gain).
- Delivered Random Forest + SHAP explainability pipeline achieving 98% prediction accuracy; surfaced top 5 regional pollution drivers on a React + Leaflet dashboard.
- Refactored React component tree with lazy-loaded map tiles; improved Lighthouse performance score from 61 → 89. Built and deployed end-to-end in 48 hours for NASA Space Apps Challenge.

Sahai: AI Mental Health & Lifestyle Ecosystem | FastAPI, Django, PostgreSQL, Redis, FAISS, Docker, WebSockets

github.com/Vaibhav20k/Sahai_Project

- Architected a RAG pipeline with FAISS vector search over curated mental health documentation; integrated LoRA fine-tuned LLM (OpenAI / Hugging Face) for empathetic, context-aware, and safe conversational responses with crisis detection guardrails
- Built real-time WebSocket chat with JWT authentication and role-based access control (students, counselors, admins); deployed with Docker containerization, NGINX reverse proxy, and GitHub Actions CI/CD
- Designed mood tracking with PHQ-9 standardized assessments, peer support forums, college-specific circles, and counselor booking system with automated crisis escalation routing to emergency contacts

Experience

Nexlayer Technologies Pvt. Ltd. | Full-Stack & Automation Intern

Jul 2025 – Oct 2025 (Remote)

- Shipped 4 client-facing React dashboards consuming internal REST APIs, reducing average client reporting turnaround from 3 days to same-day by replacing manual Excel exports with live data views.
- Integrated GitHub Actions CI/CD pipeline for the company's Node.js backend — automated lint, test, and Vercel preview deploy on every PR, cutting QA review time by 50%.
- Wrote browser-automation scripts in Python to validate UI flows across staging environments after each release, catching 11 regression bugs before they reached production.

Technical Skills

Languages: Python, JavaScript, TypeScript, C, C++

Frontend: React.js, Next.js, HTML5/CSS3, Tailwind CSS, shadcn/ui

Backend: FastAPI, Node.js, Express.js, Django, REST APIs, WebSockets

Databases: PostgreSQL, MySQL, SQLite, TimescaleDB, Redis

ML: Scikit-learn, NumPy, Pandas, Random Forest, SHAP, FAISS (RAG), Model Evaluation, Feature Engineering, Hyperparameter Tuning, Model Monitoring, A/B Testing

DevOps & Tools: Docker, Git, GitHub Actions CI/CD, NGINX, Vercel, Selenium, Microservices, API Design

Data Science: ETL, Data Pipeline, Distributed Systems, TimescaleDB, Data Warehouse, DuckDB, Arrow, Parquet